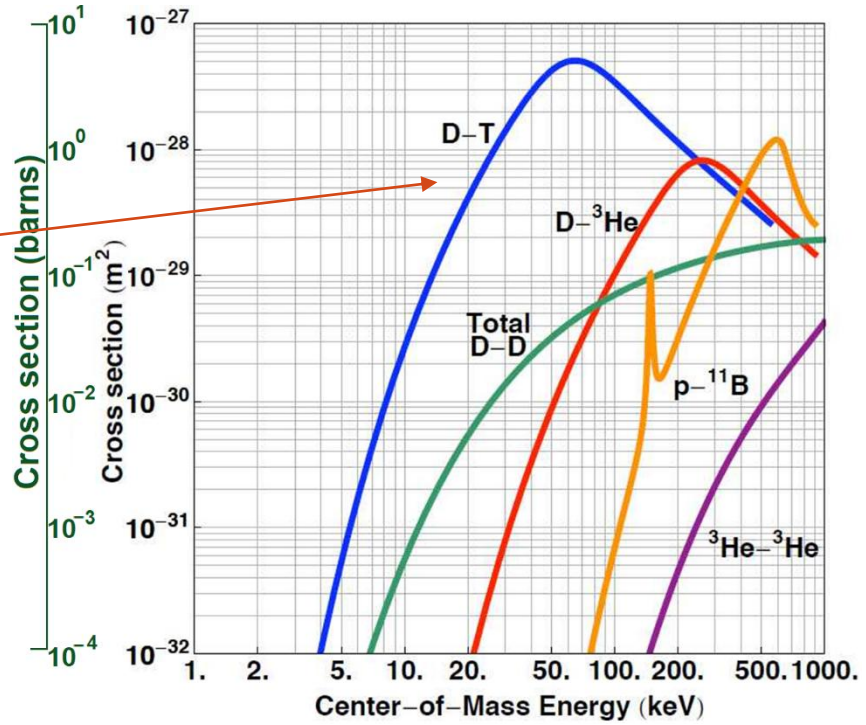
Nuclear fusion on Earth

Alternative reactions (proceeding without weak interaction):

$$D + D \rightarrow {}^{3}He(0.82\ \text{MeV}) + n(2.45\ \text{MeV})$$
$$D + D \rightarrow T(1.01\ \text{MeV}) + p(3.02\ \text{MeV})$$
$$D + {}^{3}He \rightarrow {}^{4}He(3.66\ \text{MeV}) + p(14.6\ \text{MeV})$$
$$D + T \rightarrow {}^{4}He(3.52\ \text{MeV}) + n(14.1\ \text{MeV})$$

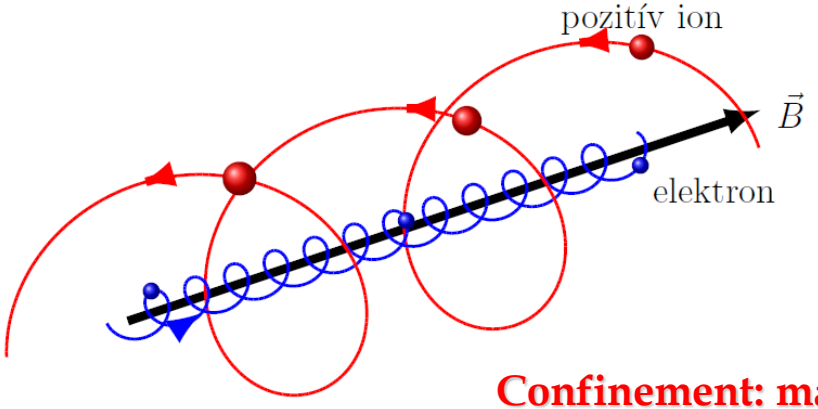
Typical ignition temperature: 4.5×10^{7} K

Plasma: plasma confinement by strong magnetic field!



Nuclear fusion on Earth

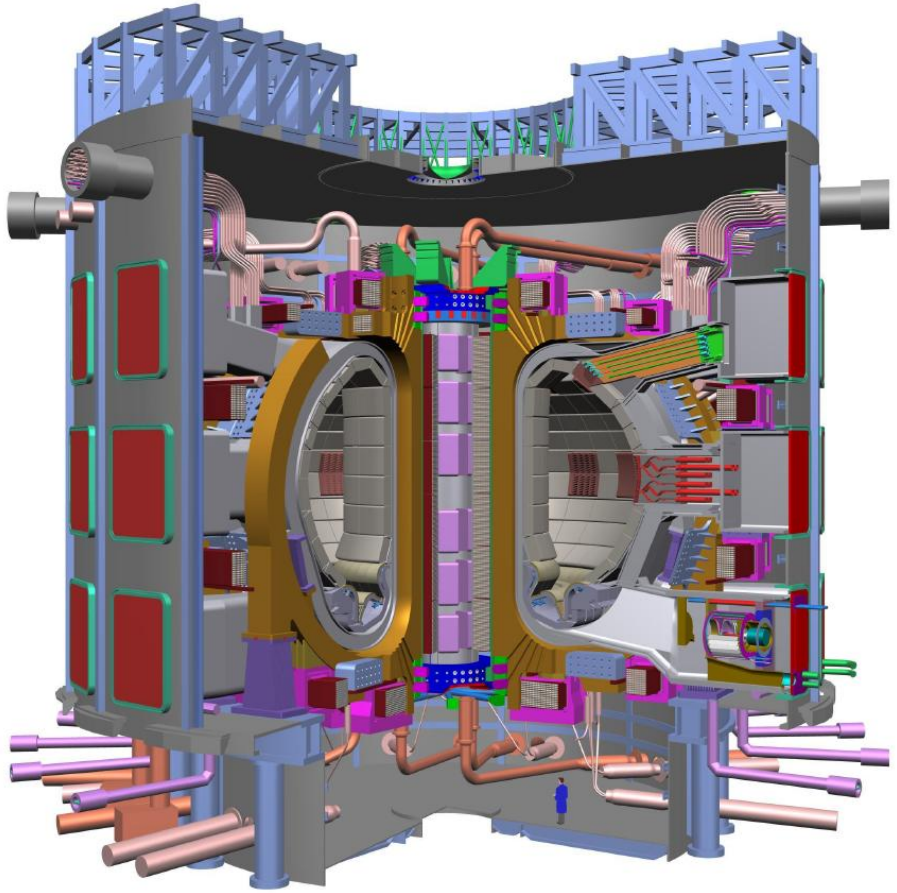
Plasma: plasma confinement by strong magnetic field!



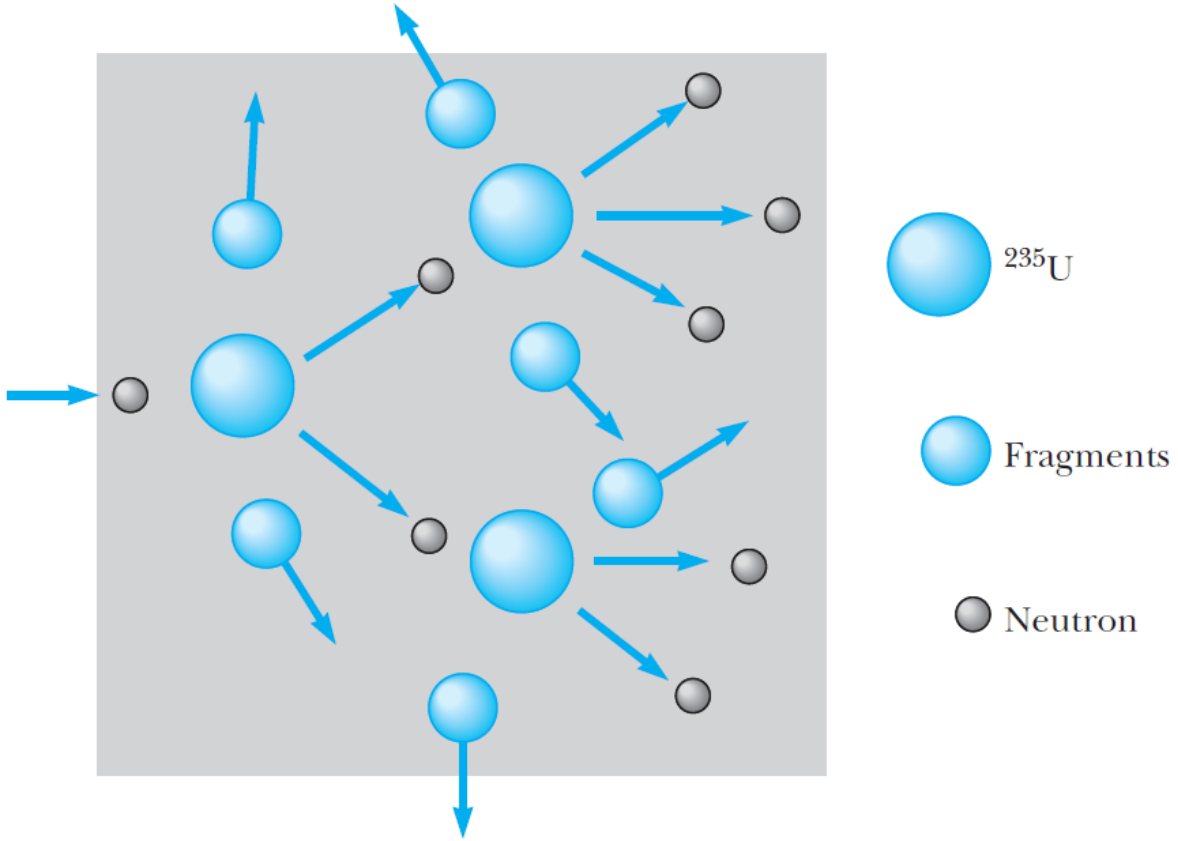
Confinement: magnetic field lines must be twisted into a helical shape inside the torus

Toroidal geometry → Stellarator OR Tokamak , ITER

Fusion group NTI!



The chain reaction



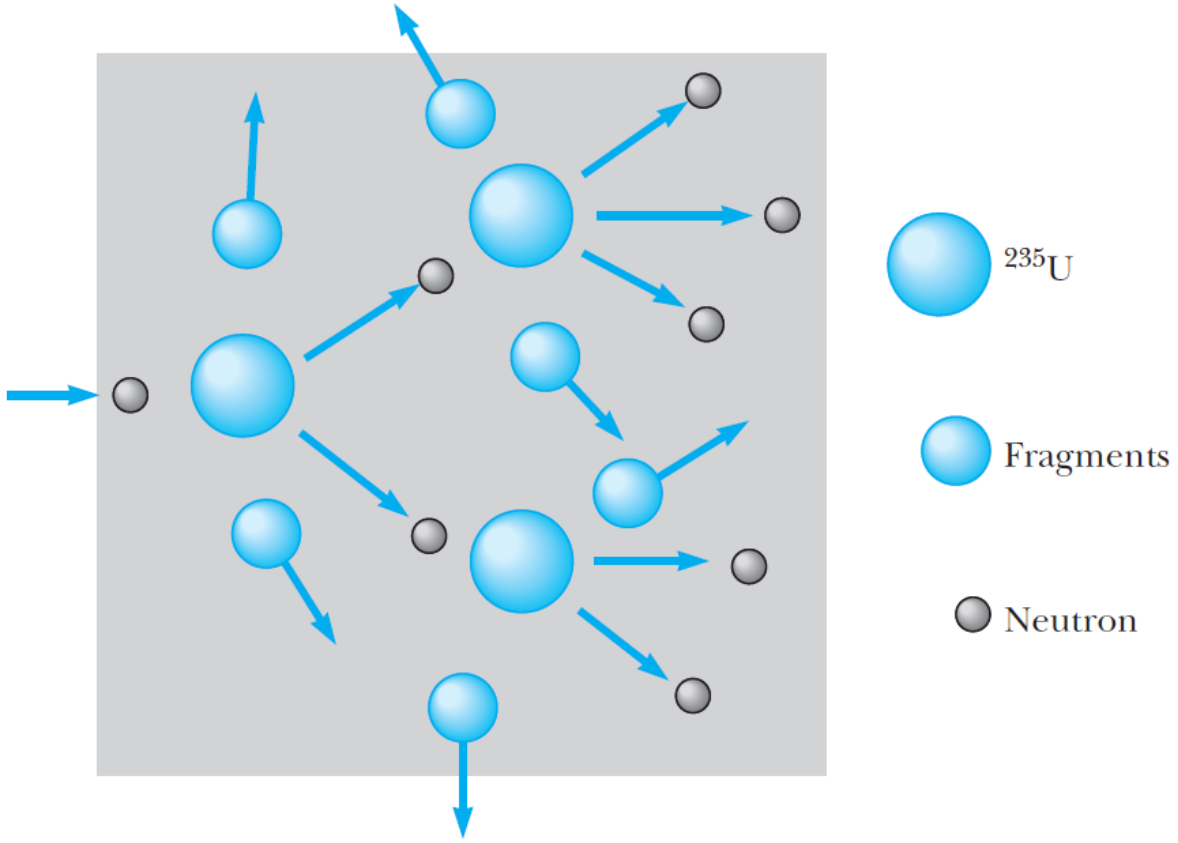
- Take a sample of fissile atoms
- Neutrons emitted during stage-1 fission could induce fission in other nuclei

- **Chain reaction**! (*Leo Szilard, 1933*)

- Condition: at least one neutron in each fission must trigger another fission

  - *Controlled chain reaction*: **power plant**
  - *Uncontrolled chain reaction*: **violent explosion** (if all atoms in 1 kg U235 would undergo fission, the energy released would be equivalent to detonating 20 kTons of TNT!) **nuclear weapon**

The chain reaction



- *Controlled chain reaction*: **power plant**
- *Uncontrolled chain reaction*: **violent explosion** (if all atoms in 1 kg U235 would undergo fission, the energy released would be equivalent to detonating 20 kTons of TNT!) **nuclear weapon**

- **Depends on the neutron reproduction**
  - **Fate of the neutrons?**
- **Depends on the fissile material**
  - **Enrichment (d)**
- **Depends on the moderator**

Basic concept of a nuclear reactor

What is the **fuel** in the reactor and what is **the fissile material**?
U235, Pu239,…
Question of *enrichment*

What is the **moderator?**
Slow neutron is more likely to induce fission

Cooling is necessary!
Heat generated by the fission reaction and also by radioactive nuclei!

Radioactivity cannot be turnded off!!

What could happen to a **neutron**?

Controlling the chain reaction!

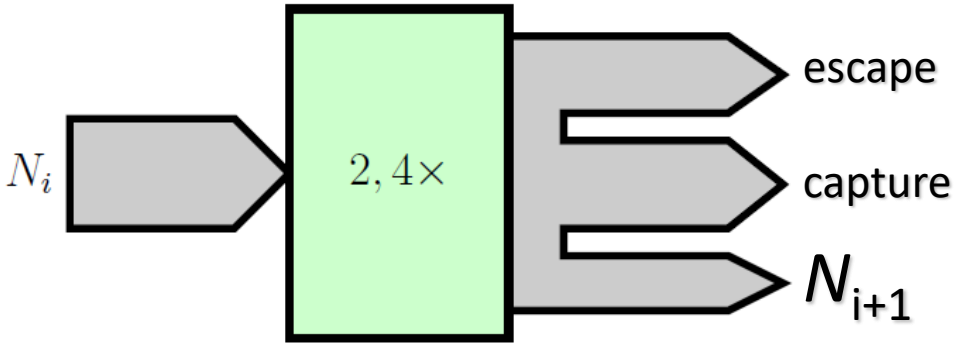
Self-sustained chain reaction

The fate of neutrons

What could happen to a neutron?
- escapes from the reactor
- (n,γ) reaction
- induces fission

Neutron „generations"

$N_1, N_2, N_3, \ldots N_i, N_{i+1}, \ldots$



N_i number of fission-inducing neutrons in the ith generation

Neutron multiplication factor:

$$k_{eff} = \frac{N_{i+1}}{N_i}$$

- $k_{eff} < 1$, chain reaction dies out („subcritical")
- $k_{eff} = 1$, chain reaction is self-sustained („critical")
- $k_{eff} > 1$, chain reaction runs away („supercritical")

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Time evolution of the chain reaction:

$$N(t) = N_0 e^{\frac{k_{eff}-1}{l}t}$$

average neutron „lifetime": time between neutron creation and absorption

$l \approx 25\ \mu s$
$k_{eff} = 1{,}001$

For such k_{eff} we have $N(t = 1s) = N_0 e^{40}$!!!!
Uncontrollable chain reaction!

The fate of neutrons

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Uncontrollable chain reaction!

New condition:

$$k_{eff} < 1 + \beta$$

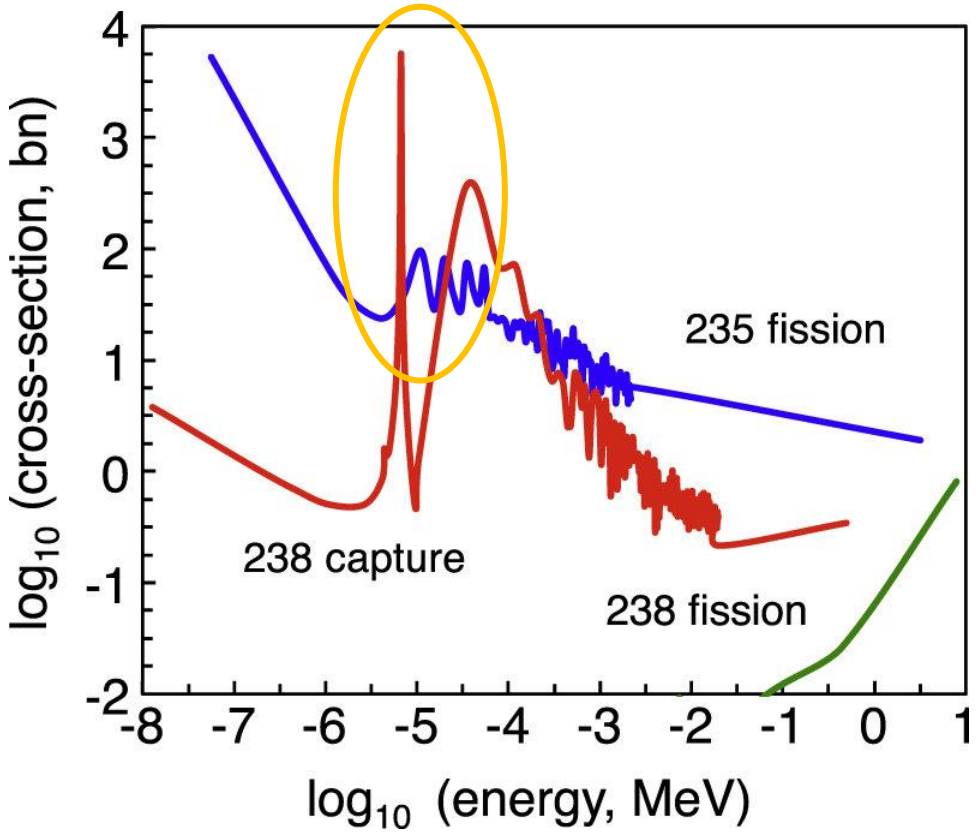
where $\beta = \sum_i \beta_i$ is the **total delayed neutron ratio**!

Delayed neutrons make it possible to sustain and control the chain reaction!

The fate of neutrons

What could happen to a neutron?

- escapes from the reactor
- **(n,γ) reaction**
- induces fission



Concentration of U-235 in the uranium fuel?
enrichment

Concentration of uranium 235

What is the **fuel** in the reactor and what is **the fissile material**?
U235, Pu239,…
Question of *enrichment*

$$d = \frac{N_{235}}{N_{235} + N_{238}}$$

- **In natural uranium today d=0.71%**
- **Depending on the moderator used, the natural uranium must be enriched at least to a few percents**

Moderators and neutron absorbers

Facilitate the chain reaction

Suitable materials:

- small mass number (high energy transfer by collision),
- high density
- high neutron-scattering cross section,
- low neutron-capture cross section.

Material	*Cross sections*
H_2O	**H: σa=0.332 barn**, σs=20.44 barn; O: σa=0.00027 barn, σs=3.76 barn.
D_2O	**D: σa=0.00053 barn**, σs=3.39 barn;
^{12}C	σs = 4.8 barn, σa =3.2 · 10−3 barn

→ 3-4% enrichment

→ natural uranium

Restrain the chain reaction

Suitable materials:

- high neutron-capture

^{10}B **σa=3840 barn**
^{113}Cd **σa=20800 barn**

control rods!

Basic concept of a nuclear reactor

